

Reproduction of Fig. 3 Context-Dependent GRN Construction for Plasmacytoma

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Introduction

Gene regulatory networks (GRNs) are graphs where genes are represented as nodes and gene-gene relationships are represented as edges. In this paper, the authors compare two related ideas:

- A **context-independent GRN**, which is a general human gene-gene interaction network. It is not specific to any disease.
- A **context-dependent GRN**, which starts from the same general interaction universe but filters and weights interactions according to how relevant the supporting PubMed papers are to a specific disease context.

For this analysis, the disease context was **plasmacytoma**, represented by the MeSH disease identifier D010954.

The aim was to reproduce the Fig. 3-style GRN construction and comparison for plasmacytoma: generate a plasmacytoma context-dependent GRN, compare it against the universal context-independent GRN, and identify central genes using PageRank.

Summary of what was rerun

The raw Fig. 2 PubMed processing was **not** rerun. In the original workflow, PubMed titles and abstracts are embedded and compared with disease queries using PubMedBERT. That requires the authors' PubMed vector database environment, so this analysis used the released precomputed PubMed-disease similarity scores from Zenodo.

The Fig. 3 GRN construction step **was rerun for plasmacytoma** using an adapted version of the original GitHub script:

Original script:

```
context-dependent-GRN/figures/fig3-0_create_context_dependent_GRN.py
```

Adapted script:

```
context-dependent-GRN/figures/fig3-0_plasmacytoma_create_context_dependent_GRN.py
```

The original script is designed to generate context-dependent GRNs for all MeSH diseases. The adapted script keeps the same core Fig. 3 GRN construction logic, but narrows the run to a single disease: plasmacytoma (D010954) in chunk 1.

Data used

The analysis used the authors' released data from Zenodo:

<https://zenodo.org/records/16416117>

The raw BioREX gene interaction file was:

context-dependent-GRN/data/fig2-3/all_human_gene_interactions_2024-12-18.csv

This file contains literature-derived human gene-gene interactions. The **context-independent GRN** was generated from all unique gene pairs in this file. This network is disease-agnostic and would be the same regardless of whether the disease being studied was plasmacytoma, breast cancer, skin cancer, or another disease.

The precomputed PubMedBERT disease-literature similarity file was:

context-dependent-GRN/data/fig2-3/All_MeSH_diseases_pmid_bert_corrs_chunk1.parquet

This file contains PubMed article relevance scores for diseases in chunk 1. Plasmacytoma is stored in this file as the column D010954.

The local MeSH annotation table identifies plasmacytoma as:

- disease label: Plasmacytoma
- MeSH ID: D010954
- chunk number: 1

How the GRNs were generated

Context-independent GRN

The context-independent GRN was generated from:

all_human_gene_interactions_2024-12-18.csv

The adapted script took all unique gene-gene pairs and treated each pair as an edge. Each edge was assigned a uniform weight of 1.0.

This means the context-independent GRN is a universal background network. It is **not plasmacytoma-specific** and does not change between diseases.

Output file:

context-dependent-GRN/data/fig2-3/plasmacytoma_fig3_adapted/context_independent_edges.csv

Plasmacytoma context-dependent GRN

The plasmacytoma context-dependent GRN was generated from two inputs:

1. all_human_gene_interactions_2024-12-18.csv
2. All_MeSH_diseases_pmid_bert_corrs_chunk1.parquet

The adapted script used the D010954 column from the PubMedBERT score file. For each gene-gene interaction, it looked up the PMID supporting that interaction, mapped the PMID to its plasmacytoma relevance score, and retained interactions with relevance score greater than 0.2.

The retained interactions were then processed using the original Fig. 3 logic:

- remove duplicate PMID-gene-pair entries
- group interactions by gene pair
- sum plasmacytoma relevance scores per gene pair
- normalize weights to counts per million
- apply $\log_{10}$ transformation

Output files:

context-dependent-GRN/data/fig2-3/plasmacytoma_fig3_adapted/plasmacytoma_context_dependent_edges.csv

context-dependent-GRN/data/fig2-3/plasmacytoma_fig3_adapted/plasmacytoma_log1p_CPM_D010954.parquet

What the original Fig. 3 script would have done

If run unchanged, fig3-0_create_context_dependent_GRN.py would process all MeSH disease chunks:

- chunk 1
- chunk 2
- chunk 3

For every disease column in those chunks, it would generate disease-specific GRN weights. The outputs would be large all-disease matrices:

- All_MeSH_diseases_log1p_CPM_chunk1.parquet
- All_MeSH_diseases_log1p_CPM_chunk2.parquet
- All_MeSH_diseases_log1p_CPM_chunk3.parquet

Each disease would be represented as one column in those matrices. For example:

- D010954 = plasmacytoma GRN
- D009101 = multiple myeloma GRN
- other MeSH IDs = other disease-specific GRNs

Therefore, the main purpose of the adaptation was to avoid regenerating every disease and instead run the same Fig. 3 construction logic only for plasmacytoma.

Code adaptations and justification

The adapted script was:

context-dependent-GRN/figures/fig3-0_plasmacytoma_create_context_dependent_GRN.py

The adaptations were:

1. Disease restricted to plasmacytoma

What changed: added `DISEASE_ID = "D010954"`.

Justification: the project only required plasmacytoma, not all MeSH diseases.

2. Chunk restricted to chunk 1

What changed: added `CHUNK = 1`.

Justification: the MeSH annotation table shows plasmacytoma is stored in chunk 1.

The original threshold (0.2) and core weighting logic were kept unchanged from fig3-0: PMID score mapping, thresholding, duplicate removal, grouping by gene pair, CPM normalization, and log1p transformation.

3. Context-independent edge list written out

What changed: saved `context_independent_edges.csv`.

Justification: the original script uses the universal pair list internally. Saving it made the baseline GRN explicit for comparison.

4. Plasmacytoma edge list written out

What changed: saved `plasmacytoma_context_dependent_edges.csv`.

Justification: this made the disease-specific GRN directly inspectable.

5. PageRank added for both networks

What changed: used `networkx.pagerank(..., weight="weight")`.

Justification: PageRank is used in the Fig. 3 notebooks for representative disease networks; this adaptation applies the same centrality idea to plasmacytoma.

6. Overlap table added

What changed: saved `top_pagerank_overlap.csv`.

Justification: this gives a clear comparison between top genes in the two networks.

The original GitHub scripts were not overwritten. The adapted script was added separately so the original repository files remained intact.

Command run

The adapted Fig. 3 script was run from the extracted GitHub repository's `figures` directory:

```
cd C:\Users\Bokan\Documents\context-dependent-GRNs\context-dependent-GRN\figures
..\..\venv\Scripts\python.exe fig3-0_plasmacytoma_create_context_dependent_GRN.py
```

The script generated outputs in:

`context-dependent-GRN/data/fig2-3/plasmacytoma_fig3_adapted/`

PageRank calculation

PageRank was calculated using NetworkX:

```
nx.pagerank(graph, weight="weight")
```

For the context-independent GRN, all edge weights were 1.0, so PageRank reflects centrality in the general background gene-gene interaction network.

For the plasmacytoma context-dependent GRN, edge weights were plasmacytoma-specific literature-derived weights, so PageRank reflects centrality in the plasmacytoma-weighted network.

The PageRank outputs were:

- context_independent_top_pagerank.csv
- plasmacytoma_context_dependent_top_pagerank.csv

Results

The adapted Fig. 3 script loaded 3,017,208 BioREX interaction rows and identified 891,603 unique gene pairs. After applying the plasmacytoma relevance threshold, 293,965 interaction rows were retained before aggregation by gene pair.

Network statistics

Network	Nodes	Edges	Density	Largest component nodes	Largest component edges
Context-independent GRN	20,824	891,603	0.004112	20,789	891,585
Plasmacytoma context-dependent GRN	13,828	153,984	0.001611	13,716	153,928

The plasmacytoma context-dependent GRN was smaller and less dense than the context-independent GRN. This is expected because disease-context weighting filters the general interaction network to interactions supported by plasmacytoma-relevant literature.

Top PageRank genes

The top 10 PageRank genes in the plasmacytoma context-dependent GRN were:

Rank	Gene	PageRank
1	AKT1	0.008405
2	TP53	0.005775
3	MAPK1	0.005445
4	MYC	0.005175
5	TGFB1	0.005070
6	NFKB1	0.005004
7	CTNNB1	0.004977
8	TNF	0.004055
9	IL6	0.003541
10	STAT3	0.003347

The top 10 PageRank genes in the context-independent GRN were:

Rank	Gene	PageRank
1	AKT1	0.004450

Rank	Gene	PageRank
2	TP53	0.003384
3	NFKB1	0.003179
4	MAPK1	0.003138
5	TNF	0.003095
6	CTNNB1	0.002857
7	TGFB1	0.002856
8	MYC	0.002397
9	IL6	0.002376
10	MTOR	0.002182

Top gene overlap

Among the top 50 PageRank genes, 39 genes were shared between the context-independent and plasmacytoma context-dependent networks.

Genes appearing in the plasmacytoma top 50 but not the context-independent top 50 were:

ABL1, BRAF, CD34, CD44, IL17A, IL2, KIT, KRAS, PIK3R1, PRNP, PROM1

Genes appearing in the context-independent top 50 but not the plasmacytoma top 50 were:

APP, CDKN1A, CXCL8, GSK3B, H3P16, IL10, JUN, NFE2L2, PPARG, RELA, TLR4

Interpretation

The context-independent GRN is a universal background network. It is identical across disease analyses because it is generated from all unique human gene-gene interactions without disease-specific weighting.

The plasmacytoma context-dependent GRN is disease-specific. It starts from the same interaction universe, but interactions are filtered and weighted using plasmacytoma-specific PubMedBERT relevance scores from the D010954 column.

Several highly ranked genes were shared between the two networks, including AKT1, TP53, MAPK1, NFKB1, MYC, IL6, and STAT3. These are broad signaling and cancer-associated genes, so it is expected that they remain central in both the general and plasmacytoma-specific networks.

However, the plasmacytoma-specific network also prioritized genes such as CD34, CD44, IL2, IL17A, KIT, and PIK3R1 within its top 50 genes. These differences show that context-dependent weighting changes gene prioritization by emphasizing interactions more strongly associated with plasmacytoma-relevant literature.

Overall, this analysis reproduced the Fig. 3-style GRN construction and comparison for plasmacytoma by minimally adapting the original GitHub Fig. 3 workflow to a single disease context.

Appendix: exact adapted analysis script

The exact final analysis script added for this project was:

context-dependent-GRN/figures/fig3-0_plasmacytoma_create_context_dependent_GRN.py

```
"""
Plasmacytoma-specific adaptation of fig3-0_create_context_dependent_GRN.py.
```

The original script loops over all MeSH disease columns in all three chunks. This adaptation keeps the same weighting logic but runs only the plasmacytoma MeSH disease ID D010954, which is stored in chunk 1.

```
import time
from pathlib import Path

import networkx as nx
import numpy as np
import pandas as pd

DISEASE_ID = "D010954"
DISEASE_NAME = "plasmacytoma"
CHUNK = 1
THRESHOLD = 0.2
TOP_N = 50

def pair_from_genes(row: pd.Series) -> str:
    return "--".join(sorted([str(row["from_gene"]), str(row["to_gene"])])))

def graph_from_edges(edges: pd.DataFrame) -> nx.Graph:
    graph = nx.Graph()
    graph.add_weighted_edges_from(
        (row.from_gene, row.to_gene, float(row.weight))
        for row in edges.itertuples(index=False)
    )
    return graph

def graph_stats(graph: nx.Graph, label: str) -> dict:
    if graph.number_of_nodes() == 0:
        largest_nodes = 0
        largest_edges = 0
    else:
        largest = graph.subgraph(max(nx.connected_components(graph), key=len))
        largest_nodes = largest.number_of_nodes()
        largest_edges = largest.number_of_edges()

    return {
        "network": label,
        "nodes": graph.number_of_nodes(),
        "edges": graph.number_of_edges(),
        "density": nx.density(graph),
        "largest_component_nodes": largest_nodes,
        "largest_component_edges": largest_edges,
    }

def pagerank_table(graph: nx.Graph, top_n: int) -> pd.DataFrame:
    pagerank = nx.pagerank(graph, weight="weight")
    return (
        pd.DataFrame(pagerank.items(), columns=["gene", "pagerank"])
        .sort_values("pagerank", ascending=False)
        .head(top_n)
        .reset_index(drop=True)
    )

def main() -> None:
    start = time.time()
    project_root = Path(__file__).resolve().parents[1]
    data_dir = project_root / "data" / "fig2-3"
    output_dir = data_dir / "plasmacytoma_fig3_adapted"
    output_dir.mkdir(parents=True, exist_ok=True)

    relations_path = data_dir / "all_human_gene_interactions_2024-12-18.csv"
    pmid_corrs_path = data_dir / f"All_MeSH_diseases_pmid_bert_corrs_chunk{CHUNK}.parquet"
```

```

print(f"Loading BioREX gene interactions: {relations_path}")
relations_df = pd.read_csv(relations_path)
relations_df["pair"] = relations_df.apply(pair_from_genes, axis=1)
all_pairs = np.array(sorted(relations_df["pair"].unique()))
print("Raw interaction rows:", len(relations_df))
print("Unique gene pairs:", len(all_pairs))

print(f"Loading Fig. 2 PubMedBERT scores: {pmid_corrs_path}")
pmid_corrs = pd.read_parquet(pmid_corrs_path, columns=["pmid", DISEASE_ID])
pmid_to_corr = pmid_corrs.set_index("pmid")[DISEASE_ID]

disease_relations_df = relations_df.copy()
disease_relations_df["corr"] = disease_relations_df["pmid"].map(pmid_to_corr)
disease_relations_df =
disease_relations_df.dropna(subset=["corr"]).reset_index(drop=True)
disease_relations_df = disease_relations_df[disease_relations_df["corr"] >
THRESHOLD].reset_index(drop=True)
print(f"[{DISEASE_ID}] Interactions after thresholding:", len(disease_relations_df))

disease_relations_df = disease_relations_df.drop_duplicates(subset=["pmid",
"pair"]).reset_index(drop=True)
disease_edges = (
disease_relations_df[["pair", "corr"]]
.groupby("pair")
.agg({"corr": "sum"})
.reset_index()
)
disease_edges["corr"] = disease_edges["corr"] / disease_edges["corr"].sum() * 1e6
disease_edges["corr"] = np.log1p(disease_edges["corr"].values)
disease_edges.columns = ["pair", "weight"]
disease_edges[["from_gene", "to_gene"]] = disease_edges["pair"].str.split("--",
expand=True)
disease_edges = disease_edges[["from_gene", "to_gene", "weight", "pair"]]

pair_to_weight = dict(zip(disease_edges["pair"], disease_edges["weight"]))
matrix_df = pd.DataFrame(
{
"pair": all_pairs,
DISEASE_ID: np.frompyfunc(lambda x: pair_to_weight.get(x, 0), 1,
1)(all_pairs).astype(np.float32),
}
)

context_independent_edges = pd.DataFrame({"pair": all_pairs})
context_independent_edges[["from_gene", "to_gene"]] =
context_independent_edges["pair"].str.split("--", expand=True)
context_independent_edges["weight"] = 1.0
context_independent_edges = context_independent_edges[["from_gene", "to_gene", "weight",
"pair"]]

print("Building graphs and calculating PageRank")
independent_graph = graph_from_edges(context_independent_edges)
dependent_graph = graph_from_edges(disease_edges)

stats = pd.DataFrame(
[
graph_stats(independent_graph, "context_independent"),
graph_stats(dependent_graph, "plasmacytoma_context_dependent"),
]
)
independent_pr = pagerank_table(independent_graph, TOP_N)
dependent_pr = pagerank_table(dependent_graph, TOP_N)

independent_top = set(independent_pr["gene"])
dependent_top = set(dependent_pr["gene"])
overlap = pd.DataFrame(
{
"shared_top_genes": ["", ".join(sorted(independent_top & dependent_top))"],
"plasmacytoma_only_top_genes": ["", ".join(sorted(dependent_top -
independent_top))"],
"context_independent_only_top_genes": ["", ".join(sorted(independent_top -
dependent_top))"],
}
)

```



```

"n_shared": [len(independent_top & dependent_top)],
"top_n": [TOP_N],
}
)

matrix_df.to_parquet(output_dir / "plasmacytoma_log1p_CPM_D010954.parquet", index=False)
context_independent_edges.to_csv(output_dir / "context_independent_edges.csv",
index=False)
disease_edges.to_csv(output_dir / "plasmacytoma_context_dependent_edges.csv",
index=False)
stats.to_csv(output_dir / "network_stats.csv", index=False)
independent_pr.to_csv(output_dir / "context_independent_top_pagerank.csv", index=False)
dependent_pr.to_csv(output_dir / "plasmacytoma_context_dependent_top_pagerank.csv",
index=False)
overlap.to_csv(output_dir / "top_pagerank_overlap.csv", index=False)

elapsed = time.time() - start
print("\nNetwork stats")
print(stats.to_string(index=False))
print(f"\nWrote results to {output_dir}")
print("Elapsed time:", time.strftime("%H:%M:%S", time.gmtime(elapsed)))

if __name__ == "__main__":
main()

```